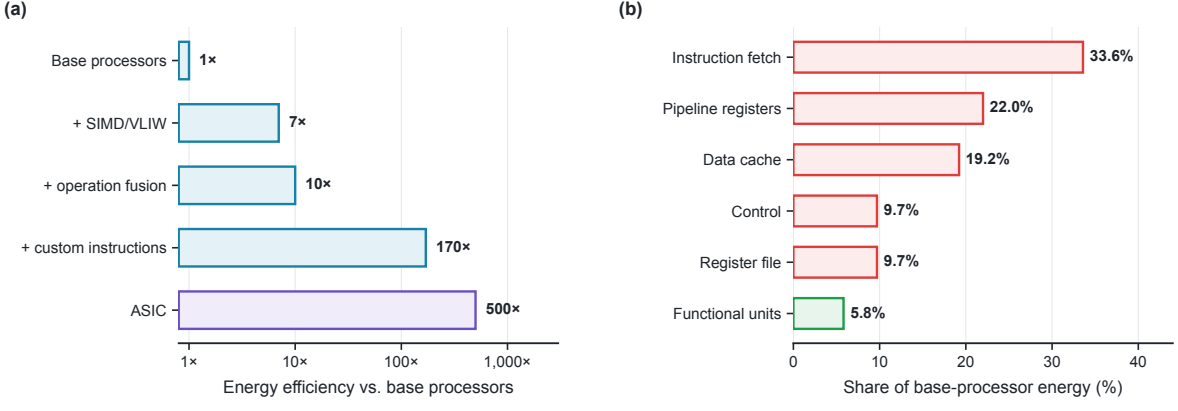
(a)
Base processors
1×
+ SIMD/VLIW
7×
+ operation fusion
10×
+ custom instructions
170×
ASIC
500×
1×
10×
100×
1,000×
Energy efficiency vs. base processors
(b)
Instruction fetch
33.6%
Pipeline registers
22.0%
Data cache
19.2%
Control
9.7%
Register file
9.7%
Functional units
5.8%
0
10
20
30
40
Share of base-processor energy (%)